

High-Q Integrated Photonics

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Abstract We review recent progress in chip-integrated high-Q resonators and discuss their applications to frequency microcombs, second-harmonic generation, and spontaneous parametric down-conversion, along with recent system-level demonstrations of high-performance microwave signal sources. ©2026 The Author

Introduction

High-Q microresonators provide access to nonlinear optical phenomena at milliwatt power levels [1]. Once discrete devices requiring specialized processing to minimize loss, microresonators are now fabricated in planar geometries, readily integrated, and in some cases produced on CMOS foundry lines. Moreover, these devices can achieve optical Q factors approaching one billion.

Following a brief overview of their history and early nonlinear demonstrations, this tutorial will highlight recent advances in devices and systems enabled by high-Q microresonator technology. Particular emphasis will be placed on efforts to bring frequency comb technology [2] to the chip scale through microcombs [3].

In addition, the recent discovery of the photogalvanic effect in silicon nitride [4–7] has opened access to second-order nonlinearities in this workhorse photonic integration platform—capabilities once limited to non-centrosymmetric dielectrics. Applications of this effect to integrated, high-coherence visible-light generation [8,9] and spontaneous parametric down-conversion [10] will also be discussed. Finally, prospects for increasing integrated optical Q factors from current limits to values as high as 10 billion will be considered [11].

Second harmonic generation in Si_3N_4

The photogalvanic effect in silicon nitride (Si_3N_4) has enabled access to second-order nonlinearities in this workhorse photonic integration platform. An example of second-harmonic generation is shown in Fig. 1a,b, where high-Q Si_3N_4 microresonators are coupled to waveguides engineered for pumping at 1064 nm and extraction of second-harmonic light at 532 nm [12].

Efficient SHG requires only the alignment of the fundamental and second-harmonic resonances. Optical poling [7] then occurs automatically to (momentum) phase-match the process. This poling mechanism writes a space-charge grating that generates a corresponding DC electric field. In combination with the intrinsic $\chi(3)$

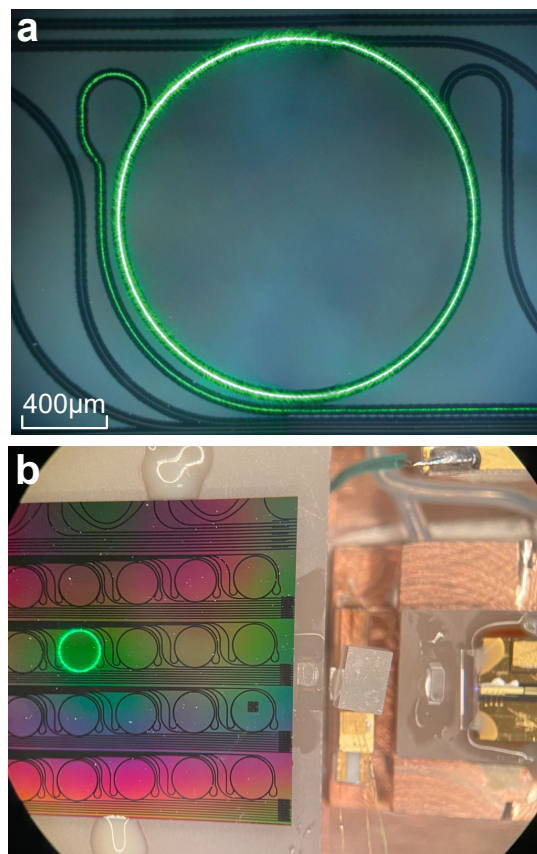


Fig. 1: (a) Photogalvanic generation of coherent green emission on chip. Photograph of a ring resonator exhibiting green (532 nm) emission during second-harmonic generation. Panel adapted from Ref. [12]. (b) Photograph of a chip containing multiple resonators, one of which is pumped at 1064 nm by a diode laser visible on the right-hand side. The laser is coupled via micro-optics and self-injection locks to the resonator, thereby experiencing a coherence enhancement. The enhanced coherence is transferred to the green emission.

nonlinearity of silicon nitride, this field induces an effective $\chi(2)$, enabling second-harmonic generation. The efficiency of the process is further enhanced by the high-Q resonators, which sustain large circulating optical intensities.

The resonators are subsequently packaged into a module that includes a diode-laser pump, coupling optics, a phase-control section, and all

required temperature-control and laser-drive electronics. A close-up photograph of the chip and diode laser is shown in Fig. 1b.

An additional aspect of this approach is its integration with self-injection locking of the pump diode lasers. In this scheme, the lasers experience a substantial boost in coherence by locking to the high-Q resonators, and this enhanced coherence is transferred to the second-harmonic emission [8,9]. Although the second-harmonic process introduces a 6 dB penalty in frequency noise, this is negligible compared to the significant noise reduction achieved through self-injection locking. Frequency-noise measurements for a 1560 nm pump laser self-injection-locked to a high-Q silicon nitride resonator are shown in Fig. 2. The short-term (high offset frequency) noise of the 780 nm second harmonic (red curve) is reduced to levels as low as 4 Hz²/Hz.

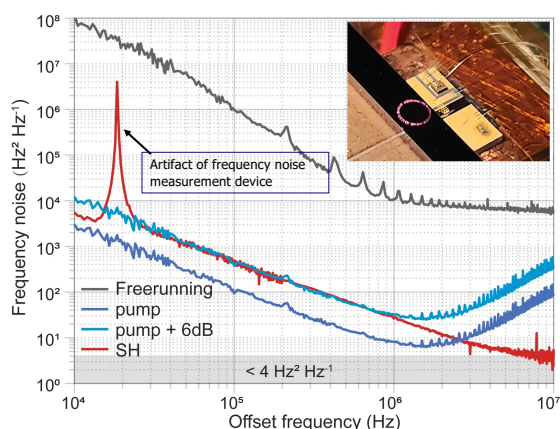


Fig. 2: Measured single-sideband frequency-noise spectra. Gray, blue, and red traces show the frequency-noise spectra of the free-running DFB laser, the self-injection-locked (SIL) pump signal (1560 nm), and the second-harmonic (SH) signal (780 nm), respectively. The light-blue trace shows the SIL pump-laser frequency noise shifted upward by 6 dB for comparison. The inset shows a photograph of the device in operation, with visible 780 nm emission from the ring resonator. Figure and data are taken from Ref. [8].

Ultra-low-loss Si₃N₄ and turn-key microcombs

Since their introduction, microcombs have advanced rapidly, addressing several key challenges in their operation [3]. Notable developments include the realization of turnkey operation [13], which greatly simplifies device startup, and the recent introduction of ultra-high-Q foundry materials [14]. These materials enable direct pumping of microcombs with semiconductor lasers and support high-volume manufacturing on CMOS fabrication lines.

In combination with new approaches to engineering mode locking in normal-dispersion waveguides via pulse-pair formation [15], these advances have enabled the first compact soliton

modules [16]. A photograph of the of a directly pumped microcomb is shown in Fig. 3a. A representative comb spectrum featuring two dispersive waves is also included in Fig. 3b. Details on turnkey module performance will be given.

These devices have recently been used to implement a form of optical frequency (OFD) division known as two-point OFD, enabling the generation of low-phase-noise microwave signals [18–20]. Performance has improved rapidly, with recent results surpassing frequency-multiplied quartz oscillators and entering the regime of sapphire oscillators [16,21].

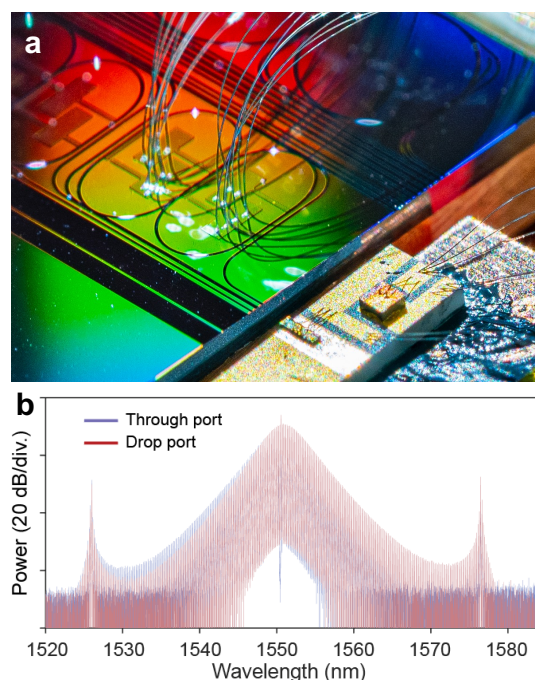


Fig. 3: (a) Turnkey soliton microcomb based on coupled high-Q racetrack resonators directly pumped by a DBR laser. The DBR laser self-injection locks to the microcomb, enabling turnkey operation [13]. Photograph adapted from Ref. [17]. (b) Optical spectrum of the coupled-racetrack microcomb, showing through-port and drop-port spectra from each racetrack. The system mode-locks via the formation of pulse pairs. Spectrum adapted from Ref. [15].

Towards fiber-like-loss on-a-chip

Despite the impressive performance of recent foundry-produced Si₃N₄ waveguides and resonators, there is precedent for even higher optical Q factors in germano-silicate waveguides used in optical fiber. Loss in telecom fiber is more than two orders of magnitude lower than the state-of-the-art losses in current foundry materials, offering the prospect of chip-integrated resonators with Q factors exceeding 10 billion. Beyond the performance gains for existing applications, such advances could enable the transfer of

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